

ApiaViz: colour and contrast preprocessing for spiking visual navigation

Preliminary simulation study

29 September 2026

Abstract. Insect-inspired navigation models often reduce an image to a small set of pixel intensities before forming a sparse neural representation. ApiaViz adds fixed colour and local-contrast processing. We compared six image-processing methods on the same four simulated routes, keeping route learning, spiking neurons, memory and steering unchanged. The visual filters were not trained. ApiaViz completed three of four routes, whereas Sobel filtering with colour completed all four. ApiaViz had lower average tracking error (11.5 versus 18.5 cm, including failures). These results show that deterministic spiking representations can support navigation, but do not establish an overall advantage for ApiaViz over simple filtering.

1 Motivation

A mushroom body is an insect brain structure involved in learning associations. Its Kenyon cells represent inputs through the activity of relatively few neurons. For navigation, a useful representation must distinguish nearby views while still allowing an animal to recognise a familiar route after moving. Ardin et al. [1] showed how sparse visual activity can support route memory. More recently, Gattaux et al. [2] combined Sobel edge filtering with directional memories and velocity control on a robot. These studies motivate a practical question: does ApiaViz provide a better visual input to the memory system than simpler processing?

2 Methods

We used four approximately 8 m routes in a reconstructed grassland with synthetic coloured landmarks (Ants 5, 8, 11 and 14, Route 2). Each condition learned from exactly the same images: nine lateral viewpoints spanning ± 20 cm at each 10 cm route interval, facing along the route. Learning recorded each sparse pattern once. No visual weights or neural connections were fitted.

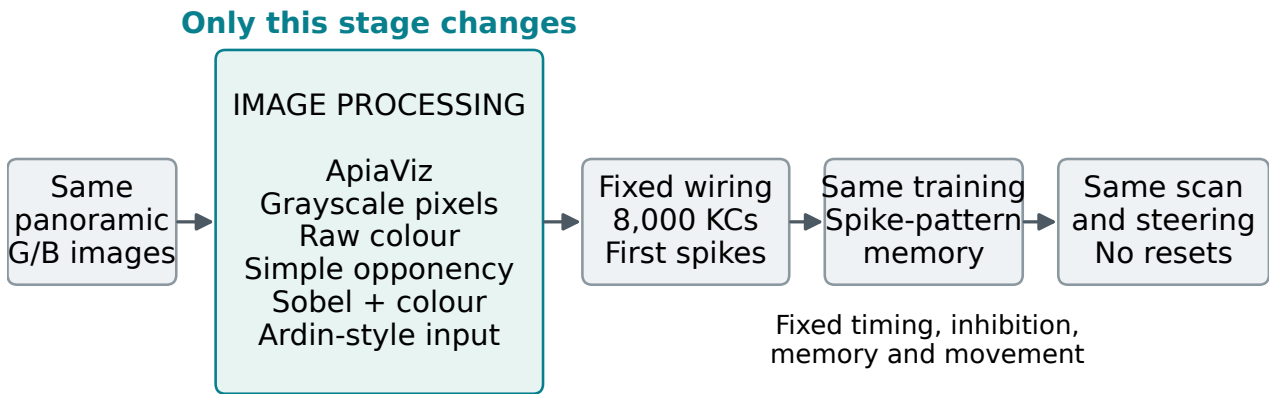


Figure 1: Controlled input substitution. Green and blue (G/B) image channels enter one of six preprocessing methods. The route-learning images, neural connections, spike parameters, memory rule and movement policy are identical across conditions. KCs: Kenyon cells.

Every method supplied five image maps, pooled to 8×64 and standardised within each of two input streams. ApiaViz used hexagonal spatial averaging, local brightness adaptation, centre-surround contrast and green-blue opponency. Controls used grayscale pixels, raw colour, simple colour differences, Sobel gradients plus colour differences, or Ardin-style inverted, locally equalised 10×36 intensity images. Single-channel inputs were copied across the maps so neither KC population was left unused. The Ardin condition applies the published image operations within our shared test pipeline.

2 Methods (continued)

All conditions used the same fixed random connections to 8,000 leaky integrate-and-fire KCs: model neurons that accumulate input until firing. Each KC sampled ten inputs and could emit one spike per 50 ms presentation. Spike times had 1 ms resolution, with a 1 ms inhibitory delay. Current gain and the inhibition trigger were fixed at the previously selected ApiaViz values for every method. We measured the resulting activity instead of forcing all outputs to have identical sparsity.

The common memory stored which KCs fired in each teaching view. Recall counted active cells shared with each stored pattern, normalised by the pattern sizes. Each route stored 693–747 patterns. The agent scanned 13 headings across $\pm 60^\circ$ and moved 10 cm towards the most familiar view. No position or heading resets were allowed. An evaluator counted arrival within 20 cm of the nest as success, with a 200-step limit. Thus navigation remained visually guided but received no external route corrections (the project’s “open-loop” condition).

3 Results

ApiaViz reached the nest on three routes; Sobel plus colour reached it on all four. Grayscale also completed three routes, while raw colour, simple opponency and Ardin-style input each completed two. ApiaViz’s mean route deviation was 11.5 cm, compared with 18.5 cm for Sobel and 33.1 cm for grayscale. Ardin-style input had a mean deviation of 66.4 cm. ApiaViz and Sobel produced similar activity levels: 4.95% of KCs for ApiaViz and 4.99% for Sobel. Their different outcomes therefore occurred at similar average sparsity.

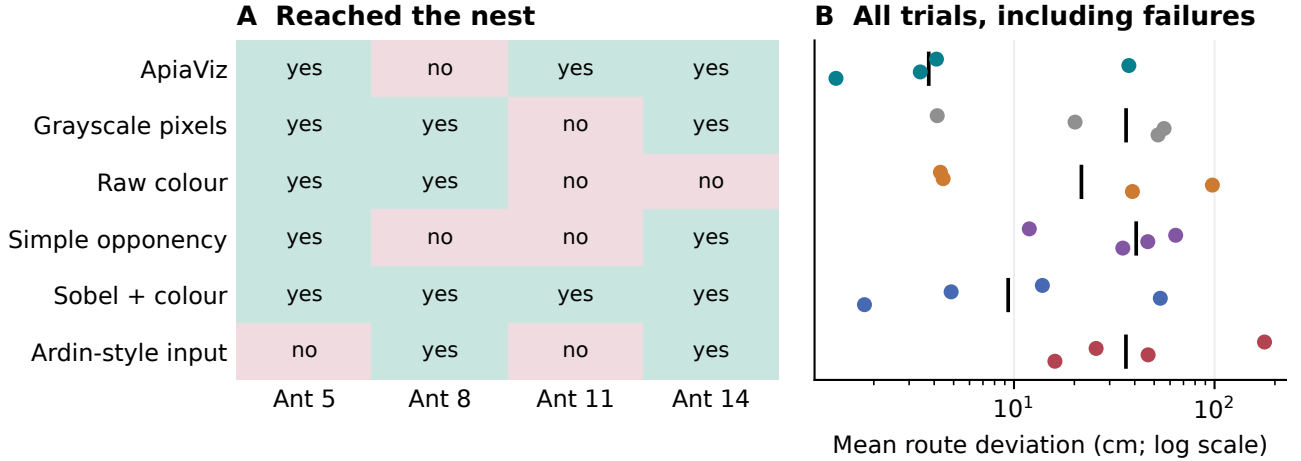


Figure 2: Open-loop comparison on the same routes. (A) Nest arrival within 200 steps. (B) Each point is one route’s mean distance from the taught centreline; black marks indicate medians. Failed trials are included. Distances are computed to the nearest route sample, spaced 10 cm apart.

Image preprocessing	Completed (of 4)	Mean deviation (cm)	Active KCs (%)
ApiaViz	3/4	11.5	4.95
Grayscale pixels	3/4	33.1	5.51
Raw colour	2/4	36.3	5.52
Simple opponency	2/4	39.3	5.36
Sobel + colour	4/4	18.5	4.99
Ardin-style input	2/4	66.4	5.55

Table 1: All four trials contribute to each mean. Activity is the fraction of KCs that fired during route learning. The same inhibition settings were used throughout; differences in actual activity arise from preprocessing.

The ApiaViz reruns reproduced their previously recorded trajectories exactly. Recorded training-image hashes, steering scores and movement traces allow the comparison to be checked independently.

4 Discussion

The comparison separates two useful outcomes: staying close to the taught route and eventually reaching the nest. ApiaViz tracked three routes closely but failed on Ant 8. Sobel completed that route and the other three, although its Ant 11 trial took 183 steps and had a mean deviation of 53.6 cm. ApiaViz may help tracking precision, but this experiment does not show that it is a more reliable navigation input than Sobel filtering.

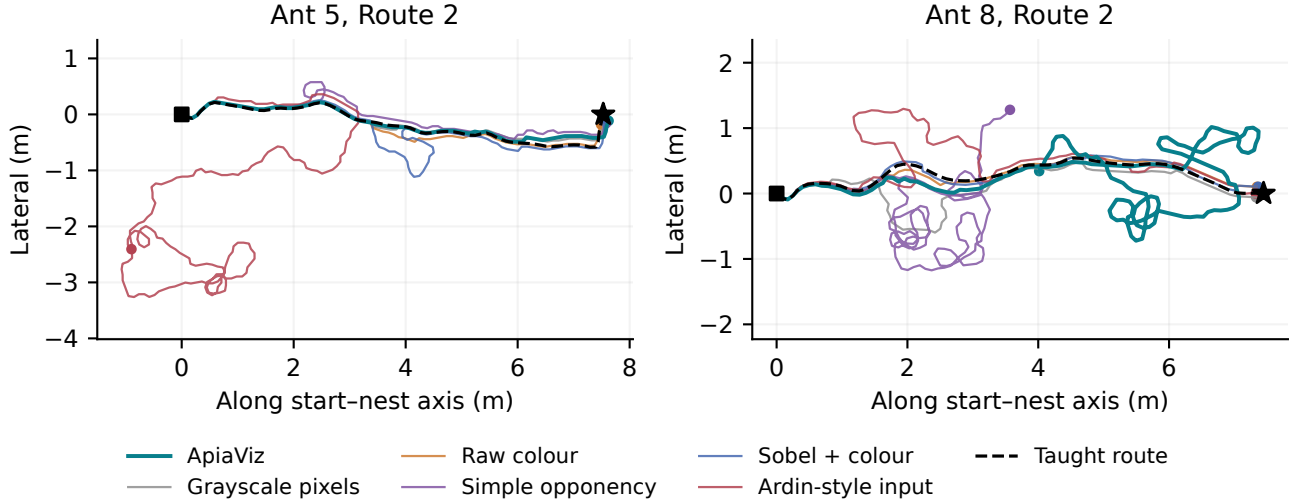


Figure 3: Examples of route following and failure. Ants 5 and 8 were chosen for illustration before observing baseline outcomes. Dashed lines show the taught routes; squares mark starts and stars mark nests. Coloured dots mark final positions. Coordinates are rotated to place the nest to the right of the start. Figure 2 and Table 1 include all four tested routes.

The matched Ardin-style condition answers a narrower question than a comparison with the original paper: what happens when its image preprocessing feeds our shared encoder and memory? Ardin et al. [1] used a different spiking circuit and corrected route departures beyond 20 cm. Its published results should therefore not be interpreted as unassisted completion scores in this experiment. Likewise, our colour-preserving Sobel condition does not reproduce the Antcar controller of Gattaux et al. [2].

This pilot covers four routes in one simulated world, with one connectivity seed and aligned starting poses. Recovery from displacement and changes in illumination require further tests. Neural settings were selected earlier for ApiaViz and then held fixed across all input substitutions. The comparison therefore describes performance at one shared setting. Template storage also uses more memory than a compact mushroom-body output circuit. The visual filters, signed input projection and memory readout use graded values; only the KCs emit spikes.

A deterministic spiking navigation system is feasible in this setting. ApiaViz reduced tracking error, while Sobel produced the most consistent completion. Recovery after drift is therefore the immediate priority for ApiaViz, followed by tests across more environments and neural wiring seeds with the same matched controls.

References

- [1] P. Ardin, F. Peng, M. Mangan, K. Lagogiannis and B. Webb (2016). Using an insect mushroom body circuit to encode route memory in complex natural environments. *PLOS Computational Biology* 12, e1004683. [doi:10.1371/journal.pcbi.1004683](https://doi.org/10.1371/journal.pcbi.1004683).
- [2] G. G. Gattaux, A. Wystrach, J. R. Serres and F. Ruffier (2025). Route-centric ant-inspired memories enable panoramic route-following in a car-like robot. *Nature Communications* 16, 8328. [doi:10.1038/s41467-025-62327-3](https://doi.org/10.1038/s41467-025-62327-3).